

AI, Human Cognition and Knowledge Collapse

Why better advice can weaken collective learning

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Acemoglu, Kong & Ozdaglar (2026) NBER WP 34910

github.com/fgalarzac/ai-04-acemoglu

Course-linked MIT: May 5, 2026 (69 pages); NBER cross-check: September 10, 2026.
Unrefereed working paper. Labels distinguish PAPER, DERIVATION and INTERPRETATION.

1. The paper and the agent's problem

PAPER: Learning jointly supplies private context and public knowledge.

AI replaces the private incentive; the public benefit goes to future cohorts.

$$\max_{e \geq 0} f(0, 0) + \Delta_G G(X) + \Delta_X G(X) G(Y) - \frac{e^\alpha}{\alpha}, \quad Y = \sigma^{-2} + \lambda_I e + \tau_A.$$

$$G(q) = 2\Phi(\sqrt{q}) - 1, \quad g(q) = \frac{\phi(\sqrt{q})}{\sqrt{q}}, \quad \boxed{e^{\alpha-1} = \Delta_X \lambda_I G(X) g(Y)} \quad (X > 0).$$

- ▶ $\Delta_I = 0$, $\Delta_X > 0$, $\Delta_G \geq 0$, $\Delta_G + \Delta_X = 1$; $\alpha > 1$.
- ▶ Medical knowledge + patient symptoms; financial knowledge + risk tolerance.
- ▶ Convex costs give a unique optimum. At $X = 0$, the optimum is $e = 0$.

PAPER: §§3.1, 3.3, 3.5. NBER uses α ; MIT uses $\varepsilon = 1/(\alpha - 1)$.

2. Observation 1: two signs, one boundary

DERIVATION, matching PAPER §3.4: for $X > 0$,

$$U_{eX} = \Delta_X \lambda_I g(X) g(Y) > 0, \quad U_{e\tau_A} = \Delta_X \lambda_I G(X) g'(Y) < 0,$$

$$g'(Y) = -\frac{1 + 1/Y}{2} g(Y) < 0.$$

- ▶ More public knowledge raises the value of learning one's own context.
- ▶ AI supplies the same private precision; diminishing returns crowd out effort.
- ▶ At $X = 0$: $e = 0$, $U_{e\tau_A} = 0$; $g(X)$ is singular at zero.

Verdict: strict interior inequalities need a boundary qualification.

Issue correction: §2 is Related Literature; the static model is in §3.

3. Main result: replenishment versus crowd-out

$$X_t \longrightarrow e_t \longrightarrow E_t = Ie_t \longrightarrow X_{t+1} = \frac{X_t + \lambda_G Ie_t}{1 + \Sigma^2(X_t + \lambda_G Ie_t)}.$$

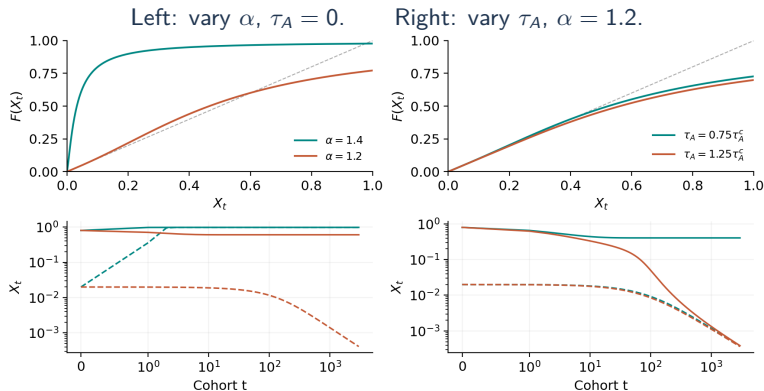
Baseline conditions: Assumption 1; power cost; atomistic short-lived Bayesian agents; Gaussian signals; fixed $I, \lambda_I, \lambda_G, \Sigma^2, \sigma^{-2} > 0$; finite $\tau_A \geq 0$; linear public learning, no synthetic input.

- ▶ $\alpha - 1 > 1/4$ ($\varepsilon < 4$): zero unstable; one positive stable state. Every $X_0 > 0$ converges to it.
- ▶ $0 < \alpha - 1 < 1/4$ ($\varepsilon > 4$): zero stable. Below τ_A^c , an unstable $\bar{X}_m$ separates collapse from $\bar{X}_h$. Above τ_A^c , all paths collapse.
- ▶ τ_A^c may be zero; $X_0 = \bar{X}_m$ stays there. At $\tau_A = \tau_A^c > 0$, Appendix A.4 retains a tangency. $\varepsilon = 4$ is a separate boundary.

PAPER intuition: chosen effort near zero is of order $X^{\varepsilon/2}$; maintenance needs X^2 .

PAPER: Lemma 2, Propositions 3 and 5; Appendix A.4–A.6, Lemmas A-1–A-4. Explained, not re-proved.

4. What I did: visualize the dynamics



$\lambda_I = \lambda_G = \Sigma^2 = \Delta_X = 1$, $I = 1000$, $\sigma^{-2} = 0.5$, $\Delta_G = \Delta_I = 0$; $X_0 = 0.02, 0.8$. 3,000 steps; time paths use log scales. Computed $\tau_A^c \approx 0.06698$. Solid paths: $X_0 = 0.8$; dashed: $X_0 = 0.02$. Full cobwebs and the parameter-specific $\alpha = 1.25$ case are in the README. Small values are not zero.

Numerical illustration of PAPER Figures 1–2, not a new result. FOC, map, crowd-out and root-count checks pass.

5. Where I did not trust the AI: welfare

Claim checked: “More accurate AI must increase welfare.”

$$\frac{d\bar{U}^+}{d\tau_A} = \underbrace{\Delta_X G(\bar{X}_h) g(\bar{Y}_h)}_{\text{direct information gain}} + \underbrace{g(\bar{X}_h) [\Delta_G + \Delta_X G(\bar{Y}_h)]}_{\text{indirect knowledge loss}} \frac{d\bar{X}_h}{d\tau_A}.$$

- ▶ Private-effort terms cancel by the envelope theorem; the public loss remains.
- ▶ Under $\sigma^{-2} \geq \sqrt{2} - 1$: single peak on the high branch, possibly at zero AI.
- ▶ An interior peak needs the direct gain to dominate initially.
- ▶ In the elastic regime, basin selection can cause an additional welfare drop.

Verdict: reject a blanket monotonicity claim; do not promise an interior optimum.

PAPER: Propositions 10–11 (high branch must exist), §§4.1–4.3; envelope calculation: DERIVATION.

6. Assumption audit and honest limits

PAPER §5 actually changes:

- ▶ §5.1: aggregation $I(\tau_A)$.
- ▶ §5.2: synthetic public precision.
- ▶ §5.3: public learning e^β .

Retained: $\Delta_I = 0$, $\Delta_X > 0$, power cost.

The unrelaxed production restriction is that context information alone has no value.

OPEN QUESTIONS: standalone context value; varying cost elasticity; contribution rewards.

INTERPRETATION: a conditional theoretical mechanism, not an empirical forecast.

Section 5 and Appendix B checked. Candidate extensions are new relative to this paper only.

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No handwritten verification is claimed yet.